

Initial Reading Notes on SABR Models for Synthetic FX Rates

Master 2 El Karoui — 2-day progress note

April 14, 2026

1. Objective of the paper

- The paper studies how to infer cross-FX option smiles from two liquid FX smiles quoted versus a common base currency.
- Main question: if $\tilde{X}_{12} = \tilde{X}_{10}/\tilde{X}_{20}$, can we obtain a reliable SABR smile for $\tilde{X}_{12}$ from the SABR models of $\tilde{X}_{10}$ and $\tilde{X}_{20}$?
- Main result: yes, through an effective lognormal SABR approximation accurate to $O(\varepsilon^2)$.

2. Global understanding

- **Main text:** financial setup, cross construction, and practical formulas for effective SABR parameters.
- **Appendix A:** general asymptotic reduction from a multi-factor stochastic-vol model to an effective one-factor forward equation.
- **Appendix B:** specialization of Appendix A to cross-FX ($n = 2$, signed loadings), giving explicit cross formulas.
- **Overall strategy:** derive marginal density PDE in one dimension, then match it to SABR's effective forward PDE.

3. Part I focused on

- I chose to focus mainly on **Appendix A**.
- Reason: it is the mathematical core that explains *why* an effective SABR emerges, not only *what* formulas to use.
- This section gives the reduction pipeline: high-dimensional dynamics $\rightarrow$ closed marginal PDE $\rightarrow$ effective SABR parameters.

4. What I understood well

- Cross rate dynamics comes naturally from Itô on $\tilde{X}_{12} = \tilde{X}_{10}/\tilde{X}_{20}$, with drift removed in the appropriate forward measure.
- The cross is driven by a signed combination of component drivers; correlation c controls cancellation/amplification of noise.

- In Appendix A, the key closure issue is clear:

$$\partial_T Q^{(0)} = \frac{1}{2} \varepsilon^2 \partial_{XX} (X^2 Q^{(2)}),$$

so one needs a constitutive relation $Q^{(2)}$ in terms of $Q^{(0)}$.

- The effective-SABR parameters are obtained by matching the resulting one-dimensional PDE with SABR's asymptotic forward PDE.

5. Key intuitions

- **Cross FX dynamics:** ratio structure implies a difference of component return shocks.
- **Effective volatility:** instantaneous variance has a quadratic form combining both vols and their correlation.
- **Why effective SABR emerges:** after asymptotic reduction, the marginal PDE has the same corrected structure as SABR ($1 + 2\varepsilon \cdot + \varepsilon^2 \cdot$).
- **Purpose of asymptotic reduction:** preserve pricing-relevant information while collapsing dimensionality.

6. What I am still working on

- Some technical details in the variable transforms used to prove the $Q^{(2)}$ - $Q^{(0)}$ relation.
- Detailed interpretation of all higher-order coefficients (especially mixed-correlation contributions).
- Full comparison between general-correlation and single-stochastic-vol cases in Appendix B.
- Calibration implications across maturities, since the mapped α_{eff} is expiry-dependent.